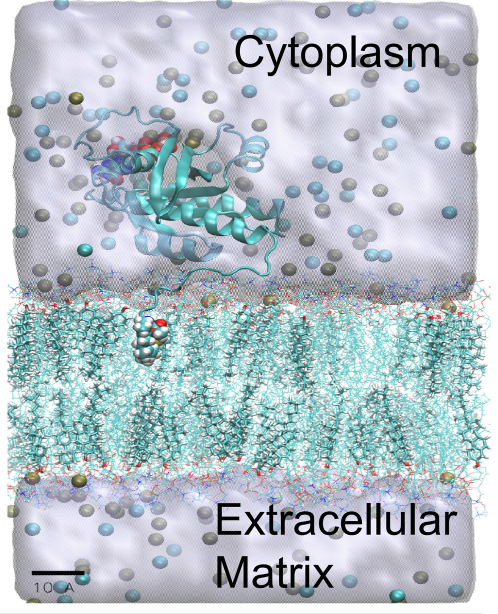

In-silico Design of a Lipid-like Inhibitor Targeting Oncogenic Peripheral Protein KRAS4B-G12D

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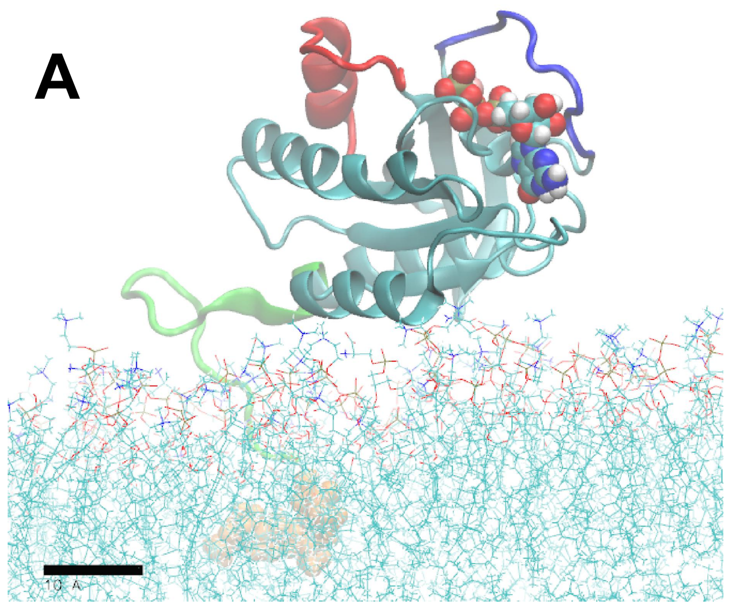
Abstract

One of the most common drivers in human cancer is the peripheral membrane protein KRAS4B, able to promote oncogenic signalling. In this work, we present an in-silico design of a lipid-like compound (LIG1) that is able to target both an oncogenic KRAS4B-G12D mutant in both GTP/GDP-bound states by tuning its orientations on membrane to block the related abnormal sianal pathways and accomodate itself in the cavity of KRAS4B's upstream porter PDEδ enzyme to impair KRAS4B enrichment on membrane. In conclusion, we report a potential inhibitor of KRAS4B-G12D with a lipid-like part attached to a especific warhead, a compound which has not yet been considered for drugs targeting RAS mutants. Our work provides a new way to target KRAS4B-G12D and can also foster drug discovery efforts for the targeting of oncogenes of the RAS family and beyond.

Introduction of KRAS4B

KRAS4B anchores into the membrane mainly through its FAR tail (orange) and establish itself throught HVR coil.



Switch I (res30-38),
Switch II (res60-76),
HVR (hypervariable region, res167-184),
Farnesyl tail (FAR, res185),
Effector-binding domain (res1-86).

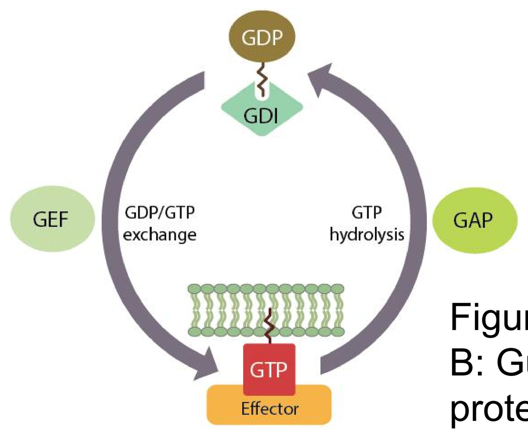
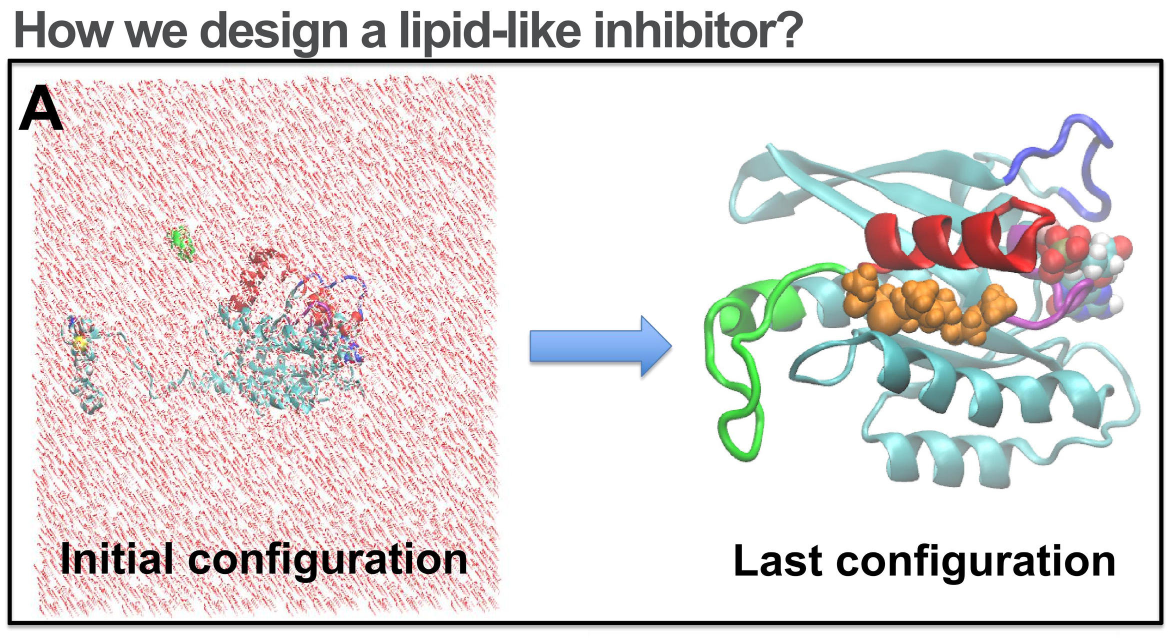


Figure 1. A: Domains of KRAS4B, B: Guanosine exchange of Ras proteins in cell.

How we design a lipid-like inhibitor?



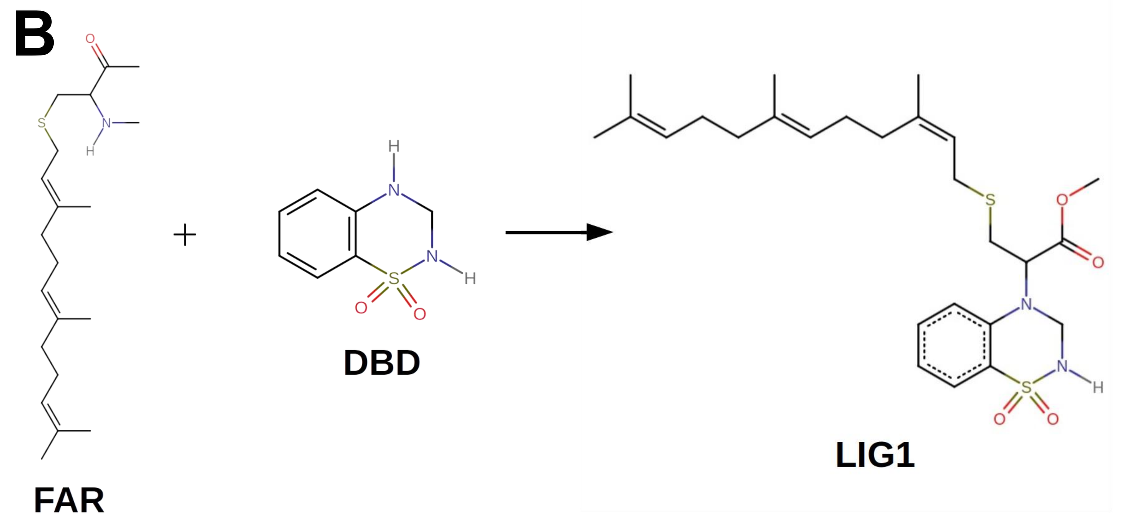


Figure 2. A: Initial and last configurations of sys2 (KRAS4B-GTP-DBD solvated in a 150 mM NaCl aqueous solution). B: Chemical strucutres of DBD, FAR, and LIG1.

Section1: LIG1 inhibits KRAS4B-G12D effectively

Preparation of systems:

Different initial configurations were generated using Chimera, and MD simulations were run using Amber22.

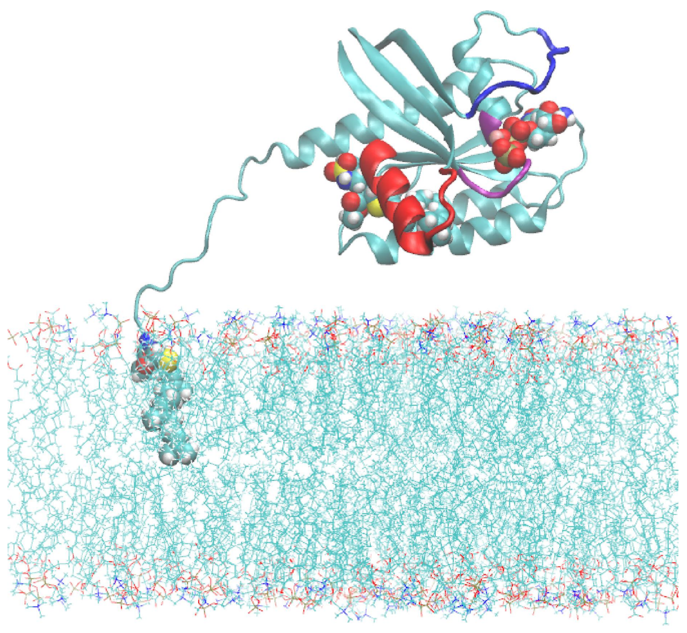


Figure 3. Illustration of initial configurations of systems, water and ions not shown.

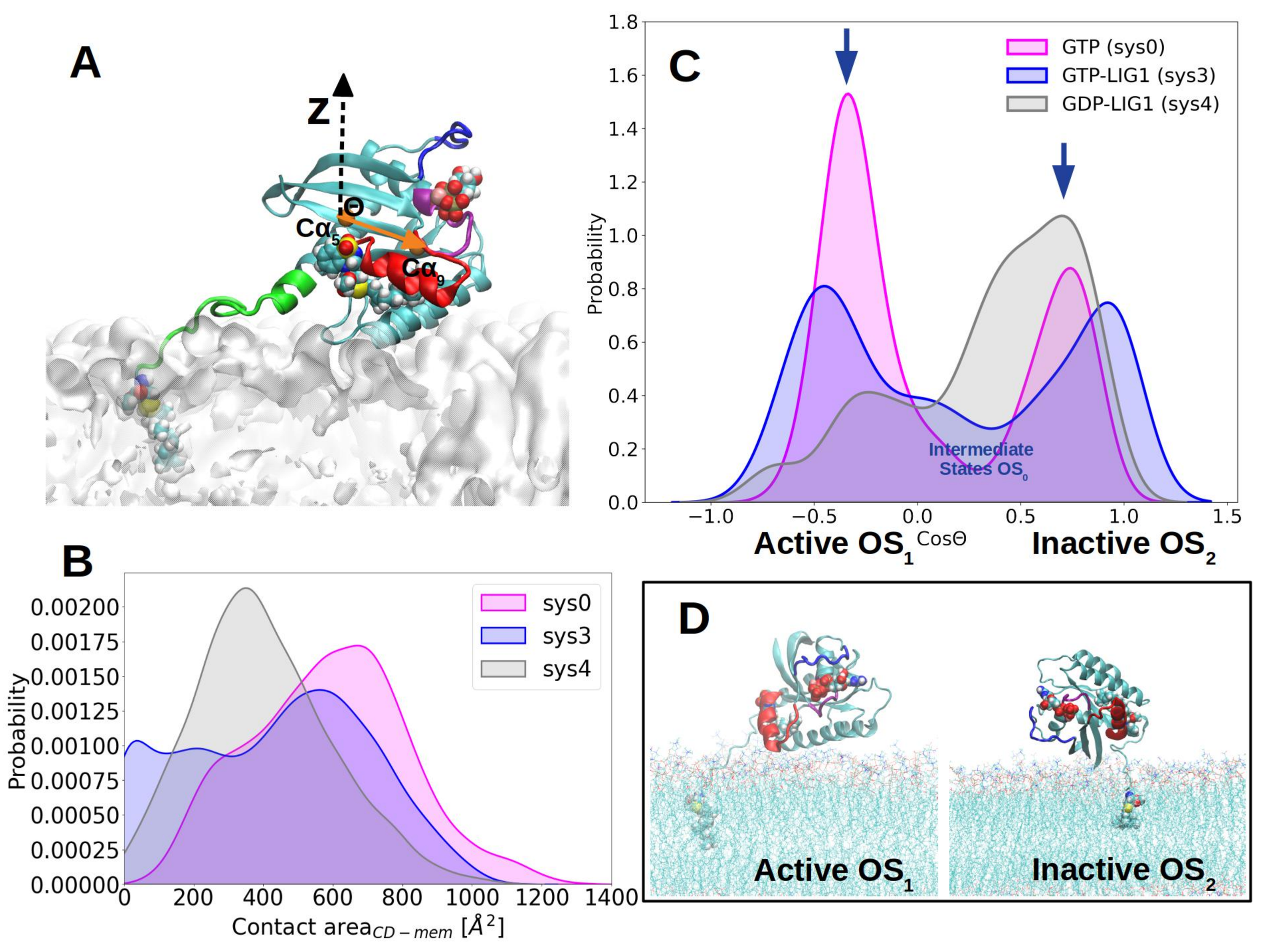


Figure 4. A: Parameter CosΘ taken to define the orientation of KRAS on membrane[1], B&C: Distribution of contact area and CosΘ, D: typical orientation states of KRAS.

Conclusion

By target into SII pocket of KRAS4B-G12D mutant, LIG1 is able to lock this Ras protein into its GDP-bound inactive orientation state on membrane, and tunes the GTP-bound mutant to shift to its inactive orientation compared with system without condering any ligand (sys0). LIG1 is a very promising KRASS4B-G12D inhibitor.

Setups of all systems studied in this work

Table 1: Detailed setups of all simulation systems considered in this work. Among them, sys1-rep and sys2-rep stand for independent production runs for sys1 and sys2, respectively.

Systems	protein	ligand	State	Environment	No. of atoms	simulation time
sys0	KRAS4B	-	GTP	PM	112618	2000 ns
sys1	KRAS4B	DBD	GTP	PM	99429	1500 ns
sys1-rep	KRAS4B	DBD	GTP	PM	99429	2500 ns
sys2	KRAS4B	DBD	GTP	solution	98560	1770 ns
sys2-rep	KRAS4B	DBD	GTP	solution	98560	1040 ns
sys3	KRAS4B	LIG1	GTP	PM	110691	4000 ns
sys4	KRAS4B	LIG1	GDP	PM	110767	3000 ns
sys5	PDEδ	LIG1	-	solution	72976	2000 ns
sys6	-	LIG1	-	PM	48808	500 ns

References

1 Prakash, P. et al., Biophysical journal 2019, 116, 179-183

Section2: LIG1 targets PDEδ

LIG1 shares the same FAR motif as full-length KRAS4B, so we have investigated whether LIG1 can act as the PDEδ inhibitor to impair KRAS4B enrichment at PM.

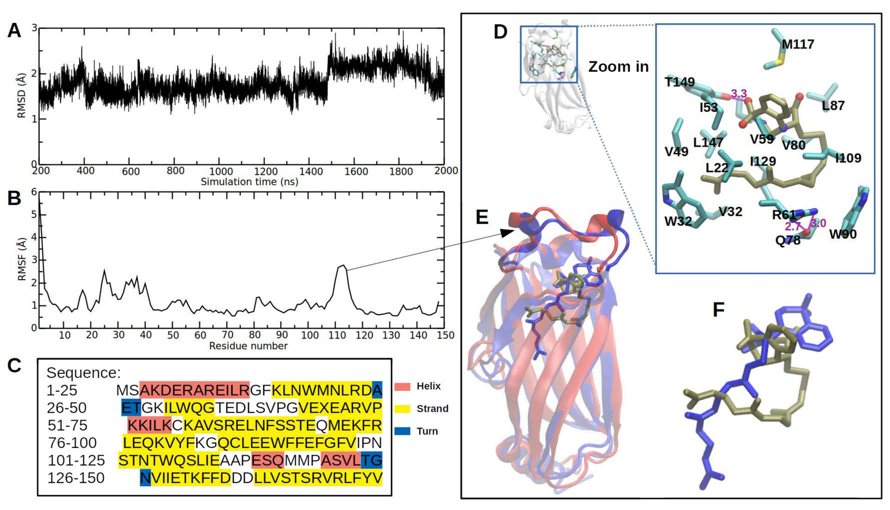


Figure 5: A, B and C: RMSD trajectories, RMSF plot and sequence of PDEδ, D: detailed hydrophobic interactions of LIG1 (tan sticks) and PDEδ.

Binding affinity:

Table 2: The corresponding binding free-energy differences ΔF between LIG1 to KRAS4B-G12D and membrane bilayer studied in this study.

System	Protein	ligand	State (orientation on PM)	ΔF (kcal/mol)
sys3	GTP	LIG1	active	80.6
sys3	GTP	LIG1	inactive	69.4
sys4	GDP	LIG1	inactive	113.5
sys5	PDEδ	LIG1	-	88.2
sys6	-	LIG1	-	63.6

Conclusion: LIG1 can bind into PDEδ with strong binding affinity, which may impair KRAS4B-G12D enrichment on the membrane.

Conclusion:

LIG1 can inhibit the abnormal signaling pathways of KRAS4B-G12D mutant in both ways.

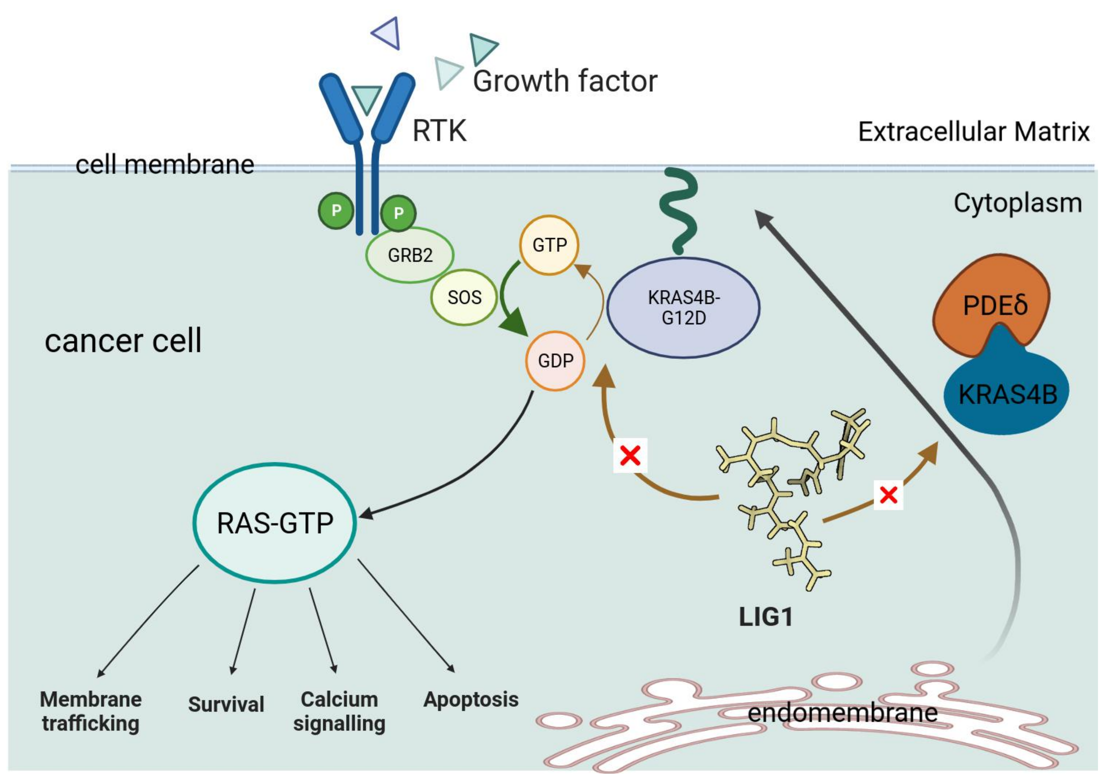


Figure 6: LIG1 is a potential KRAS4B inhibitor.

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